

Performance Statistics and Joint Optimization of CV-QKD and Classical Traffic in Dynamically Allocated Optical Networks

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Abstract We analyze the performance of classical and quantum CV-QKD systems, operating under dynamic classical traffic, assessing how classical signal statistics affects QKD performances. Furthermore, for channel-dynamics compensation, we introduce a novel gradient-descent-based approach for real-time optimization of quantum and classical throughput. ©2026 The Author(s)

Introduction

Quantum-key distribution (QKD) provides physical-layer security guaranteed by quantum physics. Most QKD implementations are transmitted over silica-core optical fibers^[1], and coexistence between quantum and classical signals on the same link is of practical interest to leverage existing networks. However, classical signals severely impair the weak quantum signal through non-linear effects^[2]. Although novel hollow-core fibers mitigate these effects^[3], standard single-mode silica fibers remain the most cost-effective solution.

Continuous-variable QKD (CV-QKD) offers the simplest integration with existing optical systems, as it utilizes weak coherent states. At CV-QKD power levels, the dominant non-linear impairments to secret key rates (SKRs) are four-wave-mixings (FWM) and spontaneous Raman scatterings (SpRS), each relevant at different regimes^[2,4].

A key aspect in practical networks, is that optical connections are established and released dynamically by the network operator. Nonetheless, all the studies that have examined coexistence penalties^[2,4,5,6,7], assumed static classical signals. Due to the sublinear scaling of throughput with signal-to-noise ratio, interference fluctuations reduce average information rates^[8,9,10], making them practically relevant. To mitigate penalties from system dynamics, future networks should jointly consider quantum and classical throughput in the operator cost function.

In this work, we evaluate statistics of the dominant coexistence interference sources using Monte Carlo simulations and our experimentally validated analytical model^[4,11] for a two-state GMCS CV-QKD signal. Results show that FWM noise exhibits larger standard deviation (STD) than SpRS interference in fully loaded C-band links. Guardbands increase SKRs by suppressing FWM while also reducing its STD. Finally, to compensate for channel dynamics, we derive a gradient-descent algorithm to jointly optimize SKRs and classical by shaping launch powers.

Interference Statistics in Dynamic Links

In silica-core fibers, non-linear interference from the dominant effects at frequency f_i is accurately modeled as^[4]

$$P_i^{\text{Int}} = \begin{cases} \left[\sum_{h \neq i} \eta_{ih} P_h L + \sum_{\substack{h,k,l \neq i \\ k=h+l-i}} \frac{\bar{\gamma}^2 P_h P_k P_l \zeta_{hl} (e^{-2\alpha L} + 1)}{4(\alpha^2 + \Delta\beta_{ihkl}^2)} \right] e^{-\alpha L}, & \text{FW,} \\ \sum_{h \neq i} \eta_{ih} P_h \frac{1 - e^{-2\alpha L}}{2\alpha}, & \text{BW,} \end{cases} \quad (1)$$

with a frequency-flat constant attenuation and effective non-linearity coefficients, α and $\bar{\gamma}$ respectively, and evenly spaced channels with index $i, h, k, l \in (1, N_{\text{ch}})$. The coefficient η_{ih} is the SpRS efficiency from f_h to f_i , and “FW” and “BW” refer to the two situations where quantum signals and classical signals propagate in the same direction or in opposite directions along the fiber, respectively. In (1), P_h are the classical launch powers, ζ_{hl} is the h^{th} channel kurtosis if $h=l$ (otherwise $\zeta_{hl} = 4$), L is the fiber length, and $\Delta\beta_{ihkl}$ is the propagation constant mismatch, given by

$$\Delta\beta_{ihkl} \approx 2\pi^2 \beta_2 (i^2 - h^2 + k^2 - l^2) \Delta f_{\text{ch}}, \quad (2)$$

where β_2 is the frequency-dependent group-velocity-dispersion parameter and Δf_{ch} the channel spacing.

Figure 1 shows co-propagating noise statistics from 1000 Monte Carlo trials using Eq. (1). Simulations consider 50 GHz-spaced 16-QAM classical signals, with Gaussian-distributed launch powers P_{Tx} ($\mu_{P_{\text{Tx}}} = -2$ dBm, $\sigma_{P_{\text{Tx}}} = 2$ dB), to simulate channel dynamics. The fiber channel parameters are $\alpha = 0.2$ dB km⁻¹, $\bar{\gamma} = \frac{8}{9} 1.3$ W km⁻¹, and $\beta_2 = -28$ ps² km⁻¹, with length $L = 25$ km.

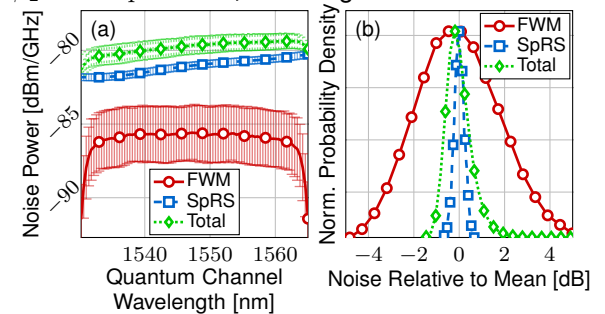


Fig. 1: (a) Noise PSD versus quantum channel from dominant sources. Error bars show ± 1 STD. (b) Normalized histograms of noise oscillations across all channels relative to their mean.

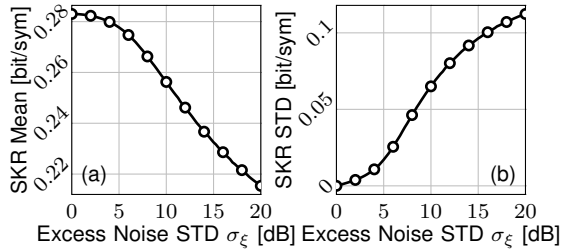


Fig. 2: (a) SKR mean and (b) STD versus excess noise fluctuations σ_ξ .

Figure 1(a) is a plot of the interference power spectral density (PSD) in a quantum channel within the C-band grid, arising from each individual source and their total. The FWM noise average decays toward edge channels (as previously reported^[12]), and exhibits significantly larger oscillations than SpRS interference.

Figure 1(b) shows the histogram of interference oscillations normalized to their maximum across all channels. The FWM interference STD exceeds SpRS, with total noise oscillations in between. This arises because FWM scales with launch power to the third power, as seen in Eq. (1), and is narrowband (sensitive mostly to adjacent channels^[2,4]), while SpRS is broadband, averaging contributions from many channels.

Statistics of SKR and Penalties

The SKRs oscillate with channel dynamics due to interference fluctuations. In stochastic channels, utilizing the average signal-to-noise ratio overestimates achievable rates due to the sublinear information-rate scaling, as previously studied for classical transmission^[9,10].

To isolate impairments solely from coexistence effects, we consider an ideal QKD receiver with ideal reverse reconciliation, and utilize the respective asymptotic formulas (Eqs. (20) and (24) of our previous work^[4]), considering interference as Gaussian and spectrally flat within the quantum channel. Practical penalties further reduce SKRs, where receiver imperfections, practical error correction algorithms, can be considered with comprehensive formulas^[13,14].

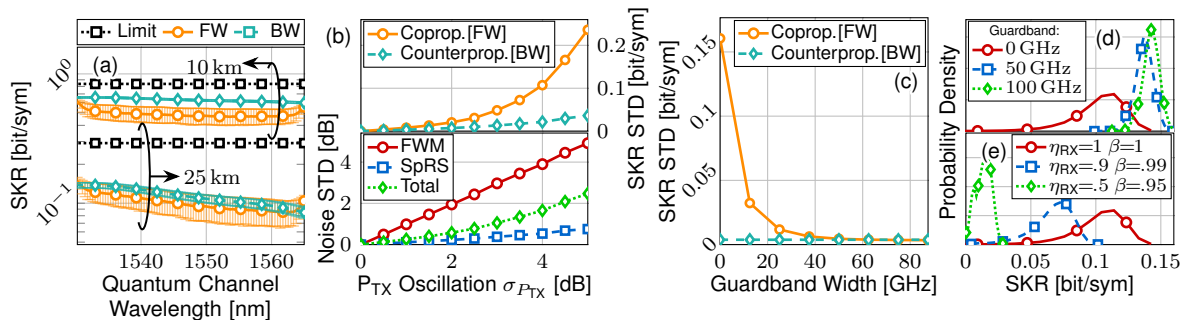


Fig. 3: (a) SKR versus quantum channel wavelength for co- (FW) and counter-propagating (BW) scenarios over two fiber lengths. Square markers show the noiseless SKR limit. Error bars show ± 1 STD. (b) SKR and noise STD versus launch power STD $\sigma_{P_{TX}}$. (c) SKR STD versus guardband width. Histograms of co-propagating SKRs for (d) different guardband widths and (e) practical receiver penalties.

Figure 2 shows asymptotic GMCS SKR mean (a) and standard deviation (b) versus excess noise STD ($\mu_\xi = -30$ dB = 10^{-3} , and 5 dB channel loss). Excess noise metrics are given in decibels as optical noise typically accumulates multiplicatively. A STD of 10 dB reflects a $\pm 1\sigma$ between $\xi \in (10^{-4}, 10^{-2})$ for the given mean value. As with classical capacity rates, average SKR decreases with increasing noise oscillations, showing that mean excess noise alone is insufficient to correctly capture QKD metrics.

Figure 3(a) shows asymptotic SKRs for a heterodyne-detected scheme versus quantum channel wavelength using the same parameters as in Fig. 1, plus counter-propagating and 10 km-link curves. Square markers show the SKR noiseless limit, giving an upper bound in the absence of interference from classical signals. At 25 km, co- and counter-propagating noise PSDs are comparable in magnitude, but co-propagation exhibits larger oscillations due to FWM. At 10 km, SKR approaches the noiseless limit with reduced interference and minimal oscillations in the counter-propagating scenario.

Figure 3(b) shows SKR and noise STD versus classical power fluctuations ($\sigma_{P_{TX}}$) over all frequency channels. Co-propagating SKR STD grows faster due to FWM. Each noise source STD increases linearly with $\sigma_{P_{TX}}$ but at different rates, while total co-propagating noise grows superlinearly.

As discussed previously in literature^[15], guardbands between quantum and classical traffic can mitigate FWM-induced penalties in co-propagating schemes. This trend is also seen in Fig. 3c, where we plot the SKR STD versus the guardband width for a channel spacing of $\Delta f_{ch} = 12.5$ GHz.

In Fig. 3(d) we plot the probability density function of the SKR for the displayed guardband values. The plot shows that co-propagating SKR distributions, considering all frequency channels, with mean normalized to the band average. A 50 GHz guardband provides significant improvement, whereas further widening yields only

marginal gain in this setup.

Figure 3(e) shows a plot of the SKR distributions under practical receiver penalties, utilizing the same channel configurations as in Fig. 3(d). These penalties reduce both average SKRs and STD, consistent with expectations.

Joint Optimization Algorithm

To compensate for dynamic oscillations in practical optical links, the network operator can shape launch powers and allocate channels according to a cost function. For links where individual classical signal powers can be shaped at the transmitter via dynamic equalization (technique typically utilized for non-linear mitigation in classical transmission^[16]), we can optimize the launch spectra for both quantum and classical throughput.

We introduce the following cost function

$$\rho = \sum_{i=1}^{N_{\text{classical}}} C_i + R \sum_{j=1}^{N_{\text{quantum}}} B_q r_j, \quad (3)$$

where $N_{\text{classical}}$ and N_{quantum} are the numbers of classical and quantum channels, C_i is the achievable rate of the i^{th} classical channel given by Shannon's formula as $C_i = B_s \log_2(1 + \text{GSNR})$, with B_s being the classical signal bandwidth and GSNR being the generalized signal-to-noise ratio, incorporating non-linear interference^[17,18,19]. In (3), B_q is the quantum signal bandwidth, and r_j is the SKR of the j^{th} quantum channel. Since classical channel capacities are typically in the order of Gbit s^{-1} while SKRs are of the order of kbit s^{-1} , R is an equivalence ratio between quantum and classical throughput.

Using the asymptotic GMCS SKR formulas, we analytically derive the formula for the derivative of the SKR with respect to excess noise for homodyne-reception as

$$\frac{dr_i}{d\xi_i} = \frac{\log_2(e)}{2T\phi(1-T)} - \frac{\log_2(e)}{2(\xi_i+1)} - \frac{\log_2\left(\frac{\omega+1}{\omega-1}\right)}{2(1-T)}, \quad (4)$$

where $T = e^{-\alpha L}$, and ϕ and ω are function of T and ξ_i defined in Eqs. (20) and (21) of our work^[4]. For heterodyne-reception, the derivative is

$$\frac{dr_i}{d\xi_i} = \frac{-\log_2(e)}{\ln(2)(\xi_i+2)} - \frac{\log_2\left(\frac{\omega+1}{\omega-1}\right)}{2(1-T)} + \frac{\log_2\left(\frac{\xi_i+2}{\xi_i+2-2T}\right)}{2T}. \quad (5)$$

The derivative of excess noise with respect to the j^{th} channel launch power P_j is

$$\frac{d\xi_i}{dP_j} = \begin{cases} \frac{e^{-\alpha L}}{2hf_i B_s} (\eta_{ij} L + \bar{\gamma}^2 \sum_{j,k,l \neq i} P_k P_l \chi_{ijkl}), & \text{FW}, \\ \frac{\eta_{ij}(1-e^{-2\alpha L})}{4\alpha h f_i B_s}, & \text{BW}, \end{cases}$$

$$\chi_{ihkl} = \begin{cases} \frac{\zeta_{jl}(e^{-2\alpha L}+1)}{2(\alpha^2 + \Delta\beta_{ijkl}^2)}, & k = j + l - i, \\ \frac{\zeta_{kl}(e^{-2\alpha L}+1)}{4(\alpha^2 + \Delta\beta_{ijkl}^2)}, & k = i + j - l. \end{cases} \quad (6)$$

The SKR gradient with respect to channel powers follows from the chain rule $\frac{dr_i}{dP_j} = \frac{dr_i}{d\xi_i} \frac{d\xi_i}{dP_j}$. Classical capacity gradients with respect to launch

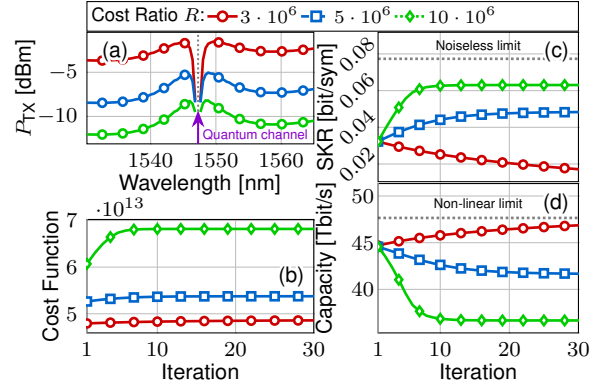


Fig. 4: (a) Optimized launch power profile for a 50 km link with distinct R values. The arrow indicates the quantum signal wavelength. (b) Cost function (3) versus iteration. Evolution of (c) SKR and (d) total capacity versus iteration.

power depend on operating regime and amplifier parameters, but follow analogously from Shannon's formula and signal-to-noise dependence. With both capacity and SKR gradients, it is straightforward deriving the gradient of Eq. (3).

Figure 4(a) shows a plot of the optimal launch power profile for a central quantum channel in a co-propagating 50 km-link configuration with bandwidth $B_q = 50$ MHz, utilizing the derived cost function gradient. We consider 6 dB pre-amplification noise figure, with target per-channel power of 0 dBm. The cost ratio R is given in the different line styles. Initial power values are $P_{Tx} = -5$ dBm and step of 10^{-19} . In the optimal profile, adjacent channels are strongly attenuated to mitigate dominant FWM interference, while near-adjacent ones reach peak power as they are positioned in the SpRS efficiency local minima.

Figure 4(b) shows the cost function versus iteration, quickly converging after only ~ 15 steps. Figures 4(c,d) show SKR (dotted lines show the noiseless limit) and total capacity evolution (dotted lines show the non-linear limit, estimated using the Gaussian-noise model^[17]). From both curves, there is a trade-off between classical and quantum throughput, which can be controlled by the operator with the R factor. Regardless of R , the algorithm quickly converges, having negligible runtime even on laptop-class systems.

Conclusions

Dynamic channel statistics in real optical networks cause SKRs oscillations, overlooked by prior co-existence studies. Using our experimentally validated coexistence model, we simulated dynamic link statistics, revealing significant FWM-induced SKRs fluctuations which can be mitigated with guardbands. To compensate for the classical traffic dynamics, we introduced a novel gradient-descent approach incorporating SKRs into the network operator's cost function with negligible computational overhead, enabling real-time joint optimization of quantum and classical throughput.

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